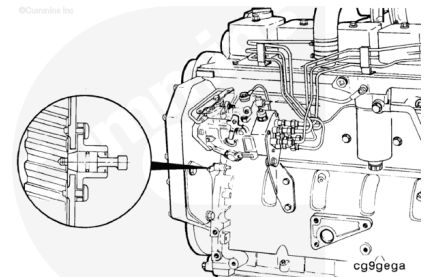


Trainings ▾

Preparatory Steps

Note : The timing pin is used to accurately locate TDC for setting the overhead. The timing pin is typically located below the fuel pump.

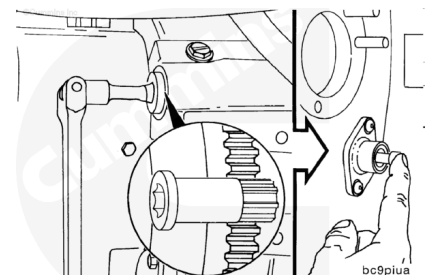
- for front gear train engines, in the front gear housing (shown)
- for rear gear train engines, in the rear gear housing (not shown)



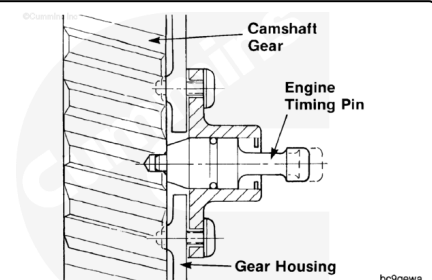
Locate top dead center (TDC) for cylinder number 1 by barring the crankshaft slowly while pressing on the engine timing pin. Barring the engine is recommended from the flywheel on the rear of the engine.

Note : Engine coolant temperature must be less than 60°C [140°F].

Using the barring tool Part Number 3824591, rotate the crankshaft slowly while pressing on the engine timing pin to locate TDC for cylinder number 1.

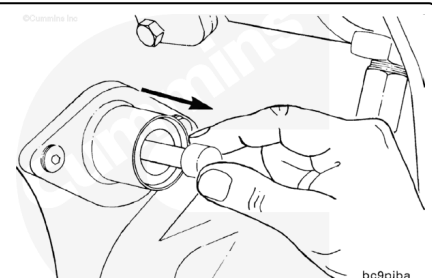


When the timing pin engages in the hole in the camshaft gear, cylinder number 1 is at TDC on the compression stroke.



⚠ CAUTION ⚠

To reduce the possibility of engine or timing pin damage, you must disengage the timing pin after locating top dead center.

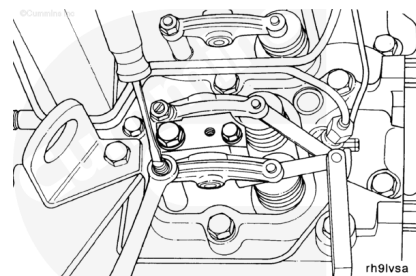


Adjust

B3.9, B5.9, and B4.5 Engines

Note : The clearance is correct when some resistance is “felt” when the feeler gauge is slipped between the valve stem and the rocker lever.

Note : Caution must be used when setting the exhaust valve lash on marine cylinder heads with rotators. The top of the valve stem is slightly recessed below the top of the valve rotator.



Measurements

	mm	in
Intake Clearance:	0.254	0.010

Measurements

	mm	in
Exhaust Clearance:	0.508	0.020

Four-Cylinder Engine Adjustment

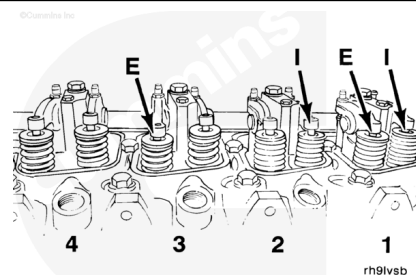
Make sure the engine is at top dead center (TDC) for cylinder number 1.

Set **only** valves indicated by the arrows (E = exhaust, I = intake). Do **not** set valves that are **not** indicated.

Holding the locknut steady with the wrench, adjust the valve clearance with the screwdriver or Allen wrench.

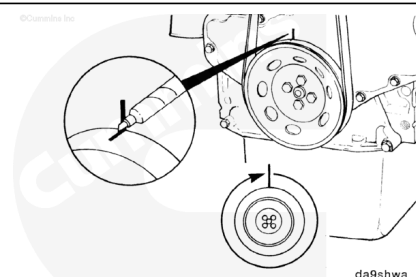
Tighten the locknut and measure the valve lash again.

Torque Value: 24 n•m [18 ft-lb]



⚠ CAUTION ⚠

To reduce the possibility of engine or pin damage, be sure the timing pin is disengaged.

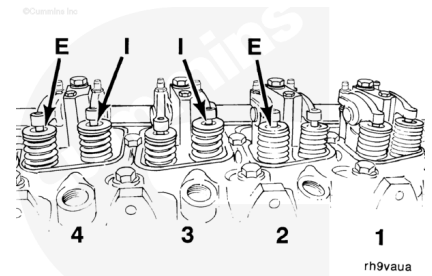


Mark the vibration damper and rotate the crankshaft 360 degrees.

Adjust the valves as indicated in the illustration.

Torque Value: 24 n•m [18 ft-lb]

Set **only** valves indicated by the arrows (E = exhaust, I = intake). Do **not** set valves that are **not** indicated.



Six-Cylinder Engine Valve Adjustment

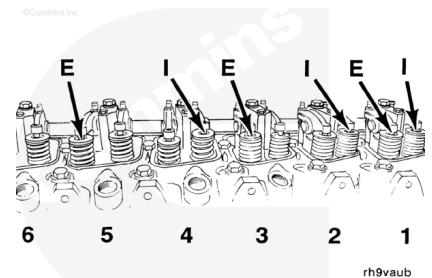
Make sure the engine is at top dead center (TDC) for cylinder number 1.

Set **only** the valves indicated by the arrows in the illustration (E = exhaust, I = intake).

Holding the locknut steady with the wrench, adjust the valve clearance with the screwdriver or Allen wrench.

Tighten the locknut, and measure the valve lash again.

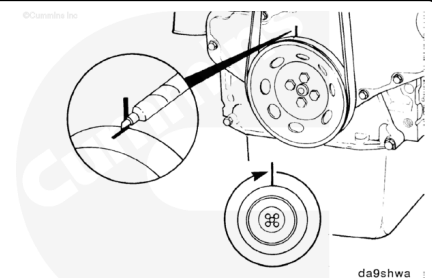
Torque Value: 24 n•m [18 ft-lb]



⚠ CAUTION ⚠

To reduce the possibility of engine or pin damage, be sure timing pin is disengaged.

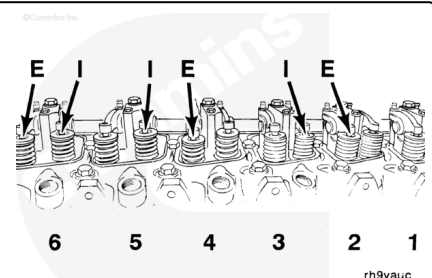
Mark the pulley, and rotate the crankshaft 360 degrees.



Adjust the valves as indicated in the illustration.

Set **only** the valves indicated by the arrows in the illustration (E = exhaust, I = intake). Do **not** set valves that are **not** indicated.

Torque Value: 24 n•m [18 ft-lb]



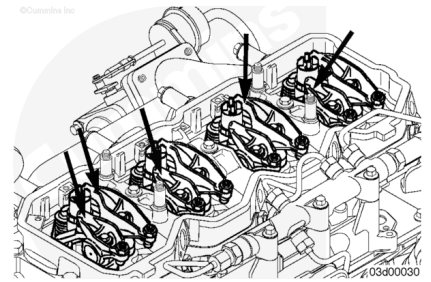
B4.5 RGT Engines

With the engine in this position, lash can be measured on the following rocker levers:

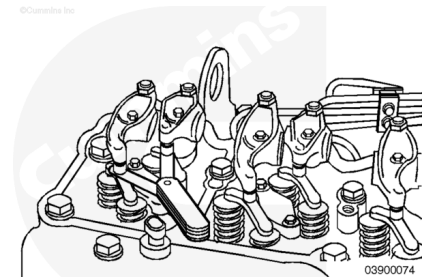
(E = exhaust, I = Intake)

Four-cylinder 1I, 1E, 2I, and 3E:

Six-cylinder 1I, 1E, 2I, 3E, 4I, and 5E.



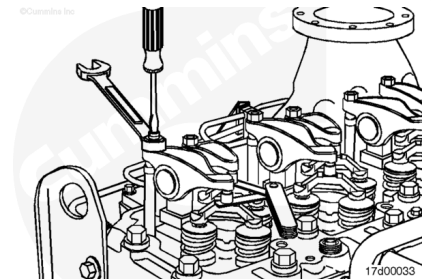
	mm		in
Intake	0.152	MIN	0.006
	0.381	MAX	0.015
Exhaust	0.381	MIN	0.015
	0.762	MAX	0.030



Note : Checking the overhead setting is usually performed as part of a troubleshooting procedure, and resetting is **not** required during checks as long as the lash measurements are within the above ranges.

Note : The clearance is correct when some resistance is “felt” when the feeler gauge is slipped between the crosshead and the rocker lever socket.

Measure lash by inserting a feeler gauge between the crosshead and the rocker lever socket. If the lash measurement is out of specification, loosen the locknut, and adjust the lash to nominal specifications.



Lash Specifications

	mm	in
Intake	0.254	0.010
Exhaust	0.508	0.020

Tighten the locknut and remeasure.

Torque Value: 24 n•m [212 in-lb]

⚠ CAUTION ⚠

To reduce the possibility of engine or pin damage, be sure the timing pin is disengaged.

Using barring tool, Part Number 3824591, rotate the crankshaft 360 degrees.

Following the same steps and specifications as previously stated, measure lash for the following rocker levers:

(E = exhaust, I = Intake)

Four-cylinder 2E, 3I, 4E, and 4I:

Six-cylinder 2E, 3I, 4E, 5I, 6I, and 6E.

Reset if out of specification.

